

CNSCC.160 – Fundamentals of Communication

Systems Tutorial Questions and Revision (Satellite Communications)

KNOWLEDGE

1. Define the terms **longitude**, **latitude**, **azimuth** angle and **elevation** angle.

Longitude: Angle east/west from the prime meridian

Latitude: Angle north/south from the equator

Azimuth angle: Angle measurement clockwise from North

Elevation angle: Angle between the pointing direction of the dish and the local horizontal plane

2. What does the acronym '**DBS**' stand for and what is a DBS service?

Direct Broadcast Satellite

3. A satellite revolves around the Earth in an **elliptical** orbit. When does the satellite reach its **maximum speed** and when does it reach its **minimum speed**?

According to Kepler's laws (2nd and 3rd), the satellite will reach its maximum speed when it is at its perigee (nearest to the Earth) and its minimum speed with it is at its apogee (farthest from the Earth).

4. Define the terms **geosynchronous** and **geostationary orbit**.

Geosynchronous orbit: Satellites rotate the Earth in 24 hours (prograde).

Geostationary orbit: Geosynchronous plus circular and zero inclination (satellite above the equator)

5. How is the **gain of an antenna** affected when (i) its beam-width increases, (ii) its diameter increases, (iii) its operating frequency reduces?

i) As the beam-width of an antenna increases, its gain decreases.

ii) As the diameter of the reflector decreases, the antenna gain increases.

iii) As the transmitting (or receiving) frequency of an antenna reduces, its gain reduces.

6. The **Inmarsat** system uses 4 geostationary satellites to provide its mobile satellite service. The competing **Iridium** system uses a network of 66 satellites in low Earth orbit instead. List three advantages of the Inmarsat system and three advantages of the Iridium system.

Inmarsat: (1) Four satellites cover almost the entire surface of the Earth, (2) Satellites are fixed relative to the ground, so accurate pointing is not needed, (3) Spot beam antennas (e.g., frequency re-use).

Iridium: (1) Small and low-power phones, (2) Small signal delay, (3) Inter-satellite links.

UNDERSTANDING

1. In a satellite TV link (e.g., from the uplink station to your home via a satellite) the antenna at the uplink station is much bigger than the antenna on your home. Why are they not both the same size?

Transmit antennas are designed to achieve a maximum power rating, whilst receive antennas differ in their noise rejection properties. The dimensions of the ground station antenna, the satellite antennas and the user antenna depend on the link budget. Larger receive antennas could be used but the noise level would increase. Downlink frequencies are usually chosen to be higher than uplink frequencies in order to further improve the gain of the receive antennas.

2. Without making any adjustments, do you think it is possible that a standard fixed (non-steerable) domestic satellite dish can receive and recover signals from more than one satellite?

Usually no. The feed-horn of a satellite dish is connected to a low-noise block (LNB) down-converter, which can only down-convert received signals (i.e., lower the frequency) of a specific polarization and frequency band. More advanced LNBs can switch between different polarizations and bands but their output provides only one band at a time. Alternatively, more than one LNBs (and feed-horns) can be used to simultaneously provide different programmes to different TV sets. In that case, the extra feed-horns are not properly centred and the received 'secondary' signals are weaker than the 'primary' signal, which is focused to the main feed-horn. Gain can be improved by using larger dishes.